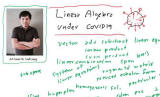


Linear Algebra under COVID19

5+1+1

Eigenvector / Eigenvalue

$\begin{bmatrix}3&1\\0&2\end{bmatrix}\begin{pmatrix}x\\y\end{pmatrix}$ $\quad B = \left\langle \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}-1\\1\end{pmatrix} \right\rangle$

$= \begin{bmatrix}1&-1\\0&1\end{bmatrix}\begin{bmatrix}3&0\\0&2\end{bmatrix}\begin{bmatrix}1&-1\\0&1\end{bmatrix}^{-1}\begin{pmatrix}x\\y\end{pmatrix}$

$Ax = \lambda x, \quad x \neq 0$

$Ax - \lambda x = 0$

$Ax - \lambda I x = 0$

$(A - \lambda I)x = 0$ singular

$\boxed{\therefore |A - \lambda I| = 0}$

$A = \begin{bmatrix}3&1\\0&2\end{bmatrix} \quad |A - \lambda I| = \left|\begin{bmatrix}3-\lambda&1\\0&2-\lambda\end{bmatrix}\right|$

$= \underbrace{(3-\lambda)(2-\lambda) = 0}_{\text{characteristic eq.}}$

spec(A) $\lambda = 3, 2$

For $\lambda = 3$

$(A - 3I)x = 0$

$A - 3I = \begin{bmatrix}3-3&1\\0&2-3\end{bmatrix} = \begin{bmatrix}0&1\\0&-1\end{bmatrix}$

$\begin{bmatrix}0&1\\0&1\end{bmatrix}$ $\quad \text{span}\left\{\begin{pmatrix}1\\0\end{pmatrix}\right\}$ eigenspace λ_3

For $\lambda = 2$

$(A - 2I)x = 0$

$A - 2I = \begin{bmatrix}3-2&1\\0&2-2\end{bmatrix} = \begin{bmatrix}1&1\\0&0\end{bmatrix}$

$\begin{bmatrix}1&1\\0&0\end{bmatrix}$ $\quad \text{span}\left\{\begin{pmatrix}-1\\1\end{pmatrix}\right\}$ eigenspace λ_2

$\begin{pmatrix}1\\0\end{pmatrix}/3 \quad \begin{pmatrix}-1\\1\end{pmatrix}/2$: Eigenvector/value

$\begin{bmatrix}0&-1\\1&0\end{bmatrix} \quad \begin{vmatrix}0-\lambda&-1\\1&0-\lambda\end{vmatrix} = \lambda^2 + 1 = 0$

$\lambda = ?$

$A = \begin{bmatrix}2&\sqrt{3}\\\sqrt{3}&4\end{bmatrix}$

$Ax = \lambda x, \quad x \neq 0$

$Ax - \lambda x = Ax - \lambda I x$

$= \underbrace{(A - \lambda I)}x = 0$

$|A - \lambda I| = 0$

$\begin{vmatrix}2-\lambda&\sqrt{3}\\\sqrt{3}&4-\lambda\end{vmatrix} = (2-\lambda)(4-\lambda) - 3$

$= \lambda^2 - 6\lambda + 8 - 3$

$= \lambda^2 - 6\lambda + 5 = (\lambda - 5)(\lambda - 1) = 0$

$\therefore \lambda = 5, 1$

$\text{spec}(A) = \{5, 1\}$

For $\lambda = 5$

$N(A - 5I) = N\left(\begin{bmatrix}2-5&\sqrt{3}\\\sqrt{3}&4-5\end{bmatrix}\right)$

$= N\left(\begin{bmatrix}-3&\sqrt{3}\\\sqrt{3}&-1\end{bmatrix}\right)$

$= N\left(\begin{bmatrix}1&-1/\sqrt{3}\end{bmatrix}\right)$

$E_5 = \text{span}\left\{\begin{pmatrix}1/\sqrt{3}\\1\end{pmatrix}\right\}$

For $\lambda = 1$

$N(A - I) = N\left(\begin{bmatrix}2-1&\sqrt{3}\\\sqrt{3}&4-1\end{bmatrix}\right) =$

$= N\left(\begin{bmatrix}1&\sqrt{3}\\\sqrt{3}&3\end{bmatrix}\right)$

$E_1 = \text{span}\left\{\begin{pmatrix}-\sqrt{3}\\1\end{pmatrix}\right\}$

$A = \begin{bmatrix}2&\sqrt{3}\\\sqrt{3}&4\end{bmatrix}$

$5 \Big/ \begin{pmatrix}1/\sqrt{3}\\1\end{pmatrix} \qquad 1 \Big/ \begin{pmatrix}-\sqrt{3}\\1\end{pmatrix}$

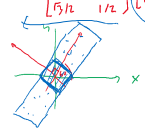
$B = \left\langle \begin{pmatrix}1/\sqrt{3}\\1\end{pmatrix}, \begin{pmatrix}-\sqrt{3}\\1\end{pmatrix} \right\rangle$

$\begin{bmatrix}5&0\\0&1\end{bmatrix}$

$\begin{bmatrix}2&\sqrt{3}\\\sqrt{3}&4\end{bmatrix} = \begin{bmatrix}1/\sqrt{3}&-\sqrt{3}\\1&1\end{bmatrix}\begin{bmatrix}5&0\\0&1\end{bmatrix}\begin{bmatrix}1/\sqrt{3}&-\sqrt{3}\\1&1\end{bmatrix}^{-1}$

$A = A^T$

$\begin{bmatrix}1/2&-\sqrt{3}/2\\\sqrt{3}/2&1/2\end{bmatrix}\begin{bmatrix}5&0\\0&1\end{bmatrix}\begin{bmatrix}1/2&\sqrt{3}/2\\-\sqrt{3}/2&1/2\end{bmatrix}$

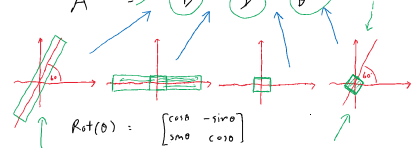


SVD

Singular Value Decomposition

$\begin{bmatrix}2&\sqrt{3}\\\sqrt{3}&4\end{bmatrix} = \begin{bmatrix}1/2&-\sqrt{3}/2\\\sqrt{3}/2&1/2\end{bmatrix}\begin{bmatrix}5&0\\0&1\end{bmatrix}\begin{bmatrix}1/2&-\sqrt{3}/2\\\sqrt{3}/2&1/2\end{bmatrix}^{-1}$

A = B D B^{-1}



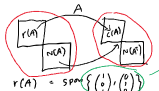
$\text{Rot}(\theta) = \begin{bmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{bmatrix}$

SVD:

$A = U \Sigma V^T$

$\begin{bmatrix}0&1&0\\1&0&1\end{bmatrix} = \underbrace{\begin{bmatrix}0&1\\1&0\end{bmatrix}}_{U}\underbrace{\begin{bmatrix}\sqrt{2}&0&0\\0&1&0\end{bmatrix}}_{\Sigma}\underbrace{\begin{bmatrix}1/\sqrt{2}&0&-1/\sqrt{2}\\0&1&0\\1/\sqrt{2}&0&1/\sqrt{2}\end{bmatrix}^T}_{V}$

$f: \mathbb{R}^3 \to \mathbb{R}^2: f(x) = Ax$



$x = \begin{pmatrix}\sqrt{2}\\1\\2\sqrt{2}\end{pmatrix}$

$r(A) = \text{span}\left\{\begin{pmatrix}1\\0\\1\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$

$N(A) = N\left(\begin{bmatrix}1&0&1\\0&1&0\end{bmatrix}\right) = \text{span}\left\{\begin{pmatrix}-1\\0\\1\end{pmatrix}\right\}$

$C(A) = \mathbb{R}^2$

$N(A^T) = \{0\}$

$\begin{bmatrix}0&1&0\\1&0&1\end{bmatrix} x = \underbrace{\begin{bmatrix}0&1\\1&0\end{bmatrix}}_{U}\underbrace{\begin{bmatrix}\sqrt{2}&0&0\\0&1&0\end{bmatrix}}_{\Sigma}\underbrace{\begin{bmatrix}1/\sqrt{2}&0&-1/\sqrt{2}\\0&1&0\\1/\sqrt{2}&0&1/\sqrt{2}\end{bmatrix}^T}_{V} x$

$x = \begin{pmatrix}\sqrt{2}\\1\\2\sqrt{2}\end{pmatrix}$

$\begin{bmatrix}1/\sqrt{2}&0&1/\sqrt{2}\\0&1&0\\-1/\sqrt{2}&0&1/\sqrt{2}\end{bmatrix}\begin{pmatrix}\sqrt{2}\\1\\2\sqrt{2}\end{pmatrix} =$

$\begin{bmatrix}\sqrt{2}&0\\0&1\end{bmatrix}\begin{pmatrix}0\\0\end{pmatrix}\begin{pmatrix}3\\1\end{pmatrix} = \begin{pmatrix}3\sqrt{2}\\1\end{pmatrix}$

$\begin{bmatrix}0&1\\1&0\end{bmatrix}\begin{pmatrix}3\sqrt{2}\\1\end{pmatrix} = \begin{pmatrix}1\\3\sqrt{2}\end{pmatrix}$

$\begin{pmatrix}\begin{pmatrix}1/\sqrt{2}\\0\\1/\sqrt{2}\end{pmatrix}\cdot\begin{pmatrix}\sqrt{2}\\1\\2\sqrt{2}\end{pmatrix}\\\begin{pmatrix}0\\1\\0\end{pmatrix}\cdot\begin{pmatrix}\sqrt{2}\\1\\2\sqrt{2}\end{pmatrix}\\\begin{pmatrix}-1/\sqrt{2}\\0\\1/\sqrt{2}\end{pmatrix}\cdot\begin{pmatrix}\sqrt{2}\\1\\2\sqrt{2}\end{pmatrix}\end{pmatrix} = \begin{pmatrix}3\\1\\1\end{pmatrix}$

$\begin{bmatrix}0&1&0\\1&0&1\end{bmatrix}\begin{pmatrix}\sqrt{2}\\1\\2\sqrt{2}\end{pmatrix} = \begin{pmatrix}1\\3\sqrt{2}\end{pmatrix}$